

The Sharp Real-Order Threshold for Harmonic Polylogarithms

Endpoint atoms, zero geometry, and certified fractional-order identities

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9 October 2026 (America/Los_Angeles)

Research continuation at a pinned repository revision

Abstract

For $a, b > 0$, let $F_{a,b}(z) = \sum_{n \geq 1} H_{n-1}^{(b)} z^n / n^a$. The existing ProveIt signed-kernel theory treats real $a \geq 1$. We prove that the exact boundary for a finite signed Hausdorff-moment representation is instead $a + b \geq 1$. Above this boundary an explicit regularized Hurwitz kernel has precisely one negative–positive sign change. On the boundary the density is replaced by a positive atom at one and a strictly negative absolutely continuous part. This distinction is necessary, not a matter of convention.

The construction extends slit-plane nonvanishing and the unique simple angular zero to the full closed region $a + b \geq 1$. On the critical line we prove strict increase of the normalized displacement $\cos \theta_{a,b}(\rho)/\rho$, through a positive integral formula for its derivative; a dual left-endpoint theorem gives strict decrease for $\text{Li}_a(z) - z$. A positive two-variable representation proves disk nonvanishing and Gaussian negativity even below the signed-measure threshold. We derive elementary critical-kernel series, complete monotonicity identities for a normalized Hurwitz expression, and strict Hankel positivity for critical harmonic defects.

The Euler evaluator extends with explicit parameter-dependent bounds. A standard-library-only rational certificate proves $\Im F_{1/10, 9/10}(i) < -0.52$, disproving extension of the old uniform constant one beyond its stated domain. Eight value enclosures and twelve angular brackets use integer-root inequalities and exact remainder bounds. The existing S_6 identity and the broader normalized-radius conjecture remain unproved here. The supplied symbolic checks and numerical diagnostics are distinguished from the written proofs and rational certificates throughout.

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1 Scope, baseline, and conventions

1.1 What is already known in the repository

The baseline is the repository *ProveIt*, revision 28357e8ca63dd78327db91d9be239d75e4462879, and in particular the canonical manuscript *Polylogarithms and their Arithmetic Bridges* and the continuation *Gaussian Polylogarithms Beyond Parity* [1, 2]. The canonical reading artifact is described by its README as a 298-page, twelve-chapter manuscript. Its current worklog marks S_4 as proved and S_6 as conjectural. Those status distinctions are retained.

The signed-kernel chapter gives integer-order moment and zero theorems. The angular continuation already extends the kernel and global nonvanishing to real $a \geq 1, b > 0$, proves several parameter comparisons at integer outer order, and proves the optimal universal Euler constant one on the domain $a \geq 1$. Re-proving those statements alone would not constitute a continuation. Our main new domain is $0 < a < 1$, especially the transition $a + b = 1$. The endpoint measure, its consequences, and the critical normalized-radius theorem are not present in the inspected baseline. The classical Hurwitz regularization and hyperbolic partial fractions used in the derivations are not claimed as new [3, 5].

Question	Inspected baseline	This article
Signed moments	Real $a \geq 1, b > 0$	Exact existence threshold $a + b \geq 1$
Critical endpoint	Not included in that domain	Required atom, exact mass, weak emergence
Slit-plane zeros	None except zero for $a \geq 1$	Same conclusion for $a + b \geq 1$
Normalized radius	Broad conjecture remains	Critical-line and dual endpoint theorems
Euler error	Optimal constant one for $a \geq 1$	Explicit extended bounds; certified scope obstruction
Below threshold	Outside signed-kernel theorem	Positive double integral, disk nonvanishing

The new results are proved below as ordinary mathematical theorems. They have not been proof-assistant formalized or independently refereed. A targeted source audit is not an exhaustive priority search: no claim is made that every identity below is absent from the entire literature. For general positive Hausdorff sequences and their analytic generating functions, see Liu and Pego [6]. Here the principal measure is signed, and its one-crossing structure must be established separately.

1.2 Definitions and boundary values

For $a, b > 0$, write

$$c_n = c_n(a, b) = \frac{H_{n-1}^{(b)}}{n^a}, \quad H_m^{(b)} = \sum_{j=1}^m j^{-b}, \quad F_{a,b}(z) = \sum_{n=1}^{\infty} c_n z^n \quad (|z| < 1). \quad (1.1)$$

In particular, $c_1 = 0$ and $c_2 = 2^{-a}$. For integer indices, $F_{a,b}(z) = \text{Li}_{a,b}(z, 1)$ in the decreasing-index convention. Set $w = a + b$ and $g_{a,b} = \text{Im } F_{a,b}(i)$, where continuation is on $\mathbb{C} \setminus [1, \infty)$. Section 5 constructs that continuation for every positive pair of parameters.

A signed Hausdorff representation means a finite real signed Borel measure μ on $[0, 1]$ such that

$$c_n = \int_{[0,1]} u^{n-1} d\mu(u), \quad n \geq 1. \quad (1.2)$$

Thus $\mu([0, 1]) = 0$. Equality of all these moments determines a finite signed measure uniquely: polynomials are uniformly dense in $C([0, 1])$, and integration against a finite measure is continuous in the uniform norm. The total variation is denoted by $\|\mu\|_{\text{TV}}$. Its two Jordan masses are equal, and we write $m(\mu) = \mu_+([0, 1]) = \mu_-([0, 1])$.

Boundary-series caution. For $w > 1$ the usual boundary series converges at every $|z| = 1, z \neq 1$. On $w = 1$ its coefficients tend to a nonzero constant, and for $w < 1$ they grow without bound. Accordingly, in those latter cases $g_{a,b}$ and boundary identities mean analytic or Abel values, not ordinary sums of a convergent series. This is proved in Section 3 and is essential for the Euler discussion.

2 The regularized Hurwitz kernel

For $s > 0$ and $T > 0$ put $k_s(T) = T^{s-1}/\Gamma(s)$, and define

$$h(t) = \frac{1}{1 - e^{-t}} - \frac{1}{t} \quad (t > 0). \quad (2.1)$$

This notation h is local to this article; it is not the positive primitive denoted by $h_{a,b}$ in the earlier angular report.

Lemma 2.1 (Elementary regularizer). *The function h increases strictly from $1/2$ to 1 , and*

$$h'(t) = \frac{1}{t^2} - \frac{1}{4 \sinh^2(t/2)} > 0. \quad (2.2)$$

For $b > 0$, $b \neq 1$, and $q > 0$,

$$\zeta(b, q) = \frac{q^{1-b}}{b-1} + \frac{1}{\Gamma(b)} \int_0^\infty e^{-qt} t^{b-1} h(t) dt. \quad (2.3)$$

At $b = 1$ the limiting formula is

$$\int_0^\infty e^{-qt} h(t) dt = \log q - \psi(q). \quad (2.4)$$

Proof. Since $2 \sinh(t/2) > t$ for $t > 0$, differentiation proves (2.2). Taylor expansion at zero and the definition at infinity give the stated limits. For $b > 1$, split the standard Hurwitz integral kernel as $(1 - e^{-t})^{-1} = t^{-1} + h(t)$. The first integral is $\Gamma(b-1)q^{1-b}$, proving (2.3) there. The remaining integral converges locally uniformly for complex $\Re b > 0$, because h is bounded. Analytic continuation in b proves the formula throughout that half-plane away from $b = 1$. This is the standard regularized Hurwitz formula, equivalently DLMF 25.11.27 [3]. Taking the constant terms in $\zeta(1 + \varepsilon, q) = \varepsilon^{-1} - \psi(q) + O(\varepsilon)$ and $q^{-\varepsilon}/\varepsilon = \varepsilon^{-1} - \log q + O(\varepsilon)$ gives (2.4). $\square$

Define the positive convolution

$$C_{a,b}(T) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T (T-t)^{a-1} t^{b-1} h(t) dt \quad (2.5)$$

$$= k_w(T) \mathbb{E}[h(TV)], \quad V \sim \text{Beta}(b, a). \quad (2.6)$$

The second parameter of this beta distribution is the outer order a . This order of parameters matters. Immediate bounds are

$$\frac{1}{2} k_w(T) < C_{a,b}(T) < k_w(T). \quad (2.7)$$

Tonelli's theorem and the gamma integral give

$$\int_0^\infty e^{-qT} C_{a,b}(T) dT = q^{-a} \left(\zeta(b, q) - \frac{q^{1-b}}{b-1} \right) \quad (b \neq 1). \quad (2.8)$$

For $b = 1$ the right side is $q^{-a}(\log q - \psi(q))$. All quantities in these formulas are finite for $q > 0$.

3 The sharp signed-measure threshold

Theorem 3.1 (Exact existence threshold). *Let $a, b > 0$. The sequence (1.1) has a finite signed Hausdorff representation if and only if $a + b \geq 1$.*

If $w > 1$, the representing measure is absolutely continuous:

$$d\mu_{a,b}(u) = K_{a,b}(-\log u) du \quad (0 < u < 1), \quad (3.1)$$

with no endpoint atoms. For $b \neq 1$ its density in logarithmic coordinates is

$$K_{a,b}(T) = \zeta(b)k_a(T) - \frac{k_{w-1}(T)}{b-1} - C_{a,b}(T). \quad (3.2)$$

For $b = 1$ it is

$$K_{a,1}(T) = k_a(T)(\gamma + \psi(a) - \log T) - C_{a,1}(T). \quad (3.3)$$

There is a unique $r \in (0, 1)$ such that the density is strictly negative on $(0, r)$ and strictly positive on $(r, 1)$.

If $w = 1$, necessarily $0 < a, b < 1$ and $a = 1 - b$. The unique measure is

$$\mu_{a,b} = \frac{1}{a}\delta_1 - \nu_{a,b}, \quad d\nu_{a,b}(u) = [-\zeta(b)k_a(-\log u) + C_{a,b}(-\log u)] du. \quad (3.4)$$

Here $\nu_{a,b}$ has strictly positive density, total mass $1/a$, and no endpoint atoms. In particular $\|\mu_{a,b}\|_{TV} = 2/a$.

3.1 Existence and the endpoint atom

Proof of the representation and its necessity. When $w > 1$, every term of (3.2) is integrable with weight $e^{-T} dT$; this follows from $a > 0$, $w - 1 > 0$ and (2.7). The logarithm in (3.3) is also integrable. By the change of variables $u = e^{-T}$,

$$\int_0^1 u^{n-1} K(-\log u) du = \int_0^\infty e^{-nT} K(T) dT.$$

Notice the factor e^{-T} in the measure: $K(T) dT$ itself is *not* its logarithmic-coordinate measure.

Using (2.8), the Laplace transform of (3.2) is

$$\zeta(b)n^{-a} - \frac{n^{-(w-1)}}{b-1} - n^{-a} \left(\zeta(b, n) - \frac{n^{1-b}}{b-1} \right) = \frac{\zeta(b) - \zeta(b, n)}{n^a} = c_n.$$

The finite Hurwitz difference is exactly $H_{n-1}^{(b)}$. At $b = 1$, differentiating the gamma integral gives

$$\int_0^\infty e^{-nT} k_a(T) \log T dT = n^{-a}(\psi(a) - \log n).$$

Together with (2.4), this makes the Laplace transform of (3.3) equal to $n^{-a}(\gamma + \psi(n)) = c_n$. In particular the total mass, the $n = 1$ case, is zero.

On the critical line, $\zeta(b) < 0$ for $0 < b < 1$. For example, the alternating Dirichlet series $\eta(b) > 0$ and $\zeta(b) = \eta(b)/(1 - 2^{1-b})$ prove this sign. Thus the density in (3.4) is strictly positive and integrable. Because $w = 1$, formula (2.8) becomes

$$\int_0^\infty e^{-nT} C_{a,b}(T) dT = n^{-a} \zeta(b, n) + \frac{1}{a}.$$

The moments of (3.4) are consequently

$$\frac{1}{a} + \zeta(b) n^{-a} - \left(n^{-a} \zeta(b, n) + \frac{1}{a} \right) = c_n.$$

The $n = 1$ identity gives $\nu([0, 1]) = 1/a$. The positive atom and the negative density are mutually singular, so the asserted total variation is exact.

It remains to exclude $w < 1$. Then $0 < b < 1$ and the integral test gives $H_{n-1}^{(b)} \sim n^{1-b}/(1-b)$, whence $c_n \sim n^{1-w}/(1-b) \rightarrow \infty$. Moments of a finite signed measure on $[0, 1]$ have absolute value at most its total variation. This is impossible. Uniqueness in the other cases follows from polynomial density, as explained in Section 1. $\square$

3.2 The one-crossing argument

Proof of the strict sign pattern when $w > 1$. Divide the proposed density by the strictly positive $k_a(T)$ and write $B(T) = K_{a,b}(T)/k_a(T)$. For $b \neq 1$, differentiation of (2.6) yields

$$\begin{aligned} B'(T) = & -\frac{\Gamma(a)}{\Gamma(w-1)} T^{b-2} \\ & -\frac{\Gamma(a)}{\Gamma(w)} T^{b-1} \mathbb{E}[bh(TV) + TVh'(TV)] < 0. \end{aligned} \quad (3.5)$$

Both terms have the displayed sign. In particular the apparent sign change of $1/(b-1)$ in the original density disappears after normalization and differentiation. Differentiation under the beta integral is valid on compact T -intervals since h and h' extend continuously to zero.

For $b > 1$, $B(T) \rightarrow \zeta(b) > 0$ as $T \downarrow 0$. For $0 < b < 1$ and $w > 1$, the term $\Gamma(a)T^{b-1}/((1-b)\Gamma(w-1))$ makes that limit $+\infty$. At infinity the term $-\Gamma(a)T^b \mathbb{E}h(TV)/\Gamma(w)$ dominates, so $B(T) \rightarrow -\infty$. This proves a unique strict crossing.

For $b = 1$,

$$B(T) = \gamma + \psi(a) - \log T - \frac{T}{a} \mathbb{E}h(TV), \quad V \sim \text{Beta}(1, a),$$

and hence

$$B'(T) = -\frac{1}{T} - \frac{1}{a} \mathbb{E}[h(TV) + TVh'(TV)] < 0. \quad (3.6)$$

Again the limits are $+\infty$ and $-\infty$. In the u coordinate, increasing T means decreasing u ; therefore the crossing has precisely the negative–positive order stated in the theorem. At the crossing $K' = k_a B' < 0$, so the crossing is transversal as well as unique. $\square$

Corollary 3.2 (Why the critical atom cannot be omitted). *On $w = 1$, every finite signed representing measure must satisfy $\mu(\{1\}) = 1/a$. No integrable density alone can represent the sequence.*

Proof. The same integral-test estimate gives $c_n \rightarrow 1/a$ on the critical line. Dominated convergence applied to u^{n-1} against the total variation of a finite measure gives $\lim c_n = \mu(\{1\})$. An absolutely continuous measure has no such atom. Equivalently, the limiting term k_0 in a formal convolution should be a Dirac mass at $T = 0$, not the zero function on $T > 0$. $\square$

Corollary 3.3 (Ordinary versus Abel boundary values). *For $w > 1$, the series defining $F_{a,b}$ converges ordinarily at every $|z| = 1, z \neq 1$ and equals its integral continuation. If $w \leq 1$, that ordinary boundary series does not converge at any such z .*

Proof. For $w > 1$, use the measure without endpoint atoms and the finite geometric sum:

$$\sum_{n=1}^N c_n z^n = \int \frac{z(1 - (zu)^N)}{1 - zu} d\mu(u).$$

For a fixed boundary point other than one the denominator is bounded away from zero uniformly in $0 \leq u \leq 1$. Dominated convergence therefore applies. If $w = 1$ the terms have modulus tending to $1/a$; if $w < 1$ their modulus grows without bound. In either case the necessary term test fails. $\square$

4 Critical defects and the emergence of the atom

Theorem 4.1 (Completely monotone critical defects). *Let $a = 1 - b \in (0, 1)$ and*

$$D_n = \frac{1}{a} - \frac{H_{n-1}^{(b)}}{n^a}, \quad n \geq 1.$$

Then $D_n = \int_0^1 u^{n-1} d\nu_{a,b}(u)$ with the strictly positive density in (3.4). If $\Delta D_n = D_n - D_{n+1}$, then $\Delta^k D_n > 0$ for every $k \geq 0, n \geq 1$. Moreover,

$$-\zeta(b)n^{-a} + \frac{1}{2n} < D_n < -\zeta(b)n^{-a} + \frac{1}{n}. \quad (4.1)$$

Every finite Hankel matrix $(D_{j+k+1})_{j,k=0}^r$ is positive definite, and $D_n D_{n+2} > D_{n+1}^2$.

Proof. Subtract the moments of (3.4) from its atomic mass. Then

$$\Delta^k D_n = \int_0^1 u^{n-1} (1-u)^k d\nu(u) > 0.$$

On $w = 1$, $k_w(T) = 1$, so (2.7) reads $1/2 < C_{a,b}(T) < 1$. Integrating these inequalities against $e^{-nT} dT$ gives (4.1). For a nonzero real vector $(v_0, \dots, v_r)$,

$$\sum_{j,k=0}^r v_j v_k D_{j+k+1} = \int_0^1 \left(\sum_{j=0}^r v_j u^j \right)^2 d\nu(u) > 0,$$

since a nonzero polynomial cannot vanish on an interval and the density is positive everywhere inside $(0, 1)$. Strict log-convexity follows either from its 2×2 minors or from strict Cauchy–Schwarz with the weight $u^{n-1} d\nu$. $\square$

The theorem also gives a continuous version. For $q > 0$,

$$D(q) = \frac{1}{a} - q^{-a}(\zeta(b) - \zeta(b, q)) = \int_0^\infty e^{-qT} [-\zeta(b)k_a(T) + C_{a,b}(T)] dT. \quad (4.2)$$

Thus $(-1)^j D^{(j)}(q) > 0$ for all integers $j \geq 0$. All derivatives are justified by exponential domination. These are inequalities for specific harmonic/Hurwitz expressions, not claims about independence of their special values.

Theorem 4.2 (Weak emergence of the critical atom). *Fix $0 < b < 1$ and let $a_\varepsilon = 1 - b + \varepsilon$ with $\varepsilon \downarrow 0$. Then $\mu_{a_\varepsilon, b} \Rightarrow \mu_{1-b, b}$ weakly on $[0, 1]$. The corresponding F converge locally uniformly on $\mathbb{C} \setminus [1, \infty)$.*

Proof. Put $M = 1/(1 - b)$. Formula (3.2) can be written

$$d\mu_{a_\varepsilon, b}(u) = M k_\varepsilon(-\log u) du - d\nu_\varepsilon(u),$$

where the density of ν_ε is $-\zeta(b)k_{a_\varepsilon}(-\log u) + C_{a_\varepsilon, b}(-\log u) > 0$. The first term has logarithmic-coordinate law M times a $\text{Gamma}(\varepsilon, 1)$ density. Its mean T is ε , so it converges weakly to $M\delta_1$ in the $u = e^{-T}$ coordinate.

The negative densities converge pointwise on $(0, 1)$ to the critical negative density. Their total masses are all M , because the positive term has mass M and the signed measure has total mass zero. The limiting negative measure also has mass M . Scheffé's lemma therefore gives convergence of the negative densities in L^1 . For completeness, this lemma follows from $\int |f_n - f| = \int f_n + \int f - 2 \int \min(f_n, f)$ and dominated convergence for $\min(f_n, f) \leq f$.

This proves weak convergence with uniformly bounded total variation. The kernels $z/(1 - zu)$ are continuous in u and uniformly bounded with their z -derivatives on compact subsets of the slit plane. A finite-net argument then upgrades pointwise convergence of their integrals to locally uniform convergence. $\square$

5 A positive two-variable representation at all positive orders

The threshold in Theorem 3.1 concerns a *one-variable finite signed measure*. It is not a threshold for analytic continuation or every positivity property.

Theorem 5.1 (Positive double kernel). *For every $a, b > 0$ and $z \in \mathbb{C} \setminus [1, \infty)$,*

$$F_{a,b}(z) = z^2 \int_{0 < y < x < 1} \frac{W_{a,b}(x, y)}{(1 - xz)(1 - yz)} dy dx, \quad (5.1)$$

where

$$W_{a,b}(x, y) = \frac{(-\log x)^{a-1} (\log(x/y))^{b-1}}{\Gamma(a)\Gamma(b)}. \quad (5.2)$$

The weight is strictly positive and has total mass 2^{-a} .

Proof. For $|z| < 1$, expand the two denominators. Substitution $y = xv$ gives, for nonnegative integers j, k ,

$$\int_{0 < y < x < 1} W_{a,b}(x, y) x^j y^k dy dx = \frac{1}{(j + k + 2)^a (k + 1)^b}.$$

Summing over $j + k = n - 2$ yields c_n . Absolute convergence justifies the rearrangement, and taking $j = k = 0$ gives the total mass. On each compact subset of $\mathbb{C} \setminus [1, \infty)$, both denominators are bounded away from zero uniformly in $x, y \in [0, 1]$. The integral is consequently holomorphic there and supplies the stated continuation. $\square$

Corollary 5.2 (Disk nonvanishing below and above the threshold). *For all $a, b > 0$, $F_{a,b}$ has no nonzero zeros in $\{z : |z| \leq 1, z \neq 1\}$. More precisely, when $\operatorname{Im} z > 0$ and $|z| \leq 1$,*

$$\operatorname{Im} \frac{F_{a,b}(z)}{z} > 0. \quad (5.3)$$

At any upper-half-disk zero of $\operatorname{Im} F_{a,b}(z)$, the value $F_{a,b}(z)$ itself is strictly negative real.

Proof. A direct calculation gives

$$\operatorname{Im} \frac{z}{(1-xz)(1-yz)} = \frac{(\operatorname{Im} z)(1-xy|z|^2)}{|1-xz|^2|1-yz|^2} > 0$$

for $0 < y < x < 1$, proving (5.3). The lower half follows by conjugation. On the real interval $z < 1$, the integral $F(z)/z^2$ is strictly positive. Finally, if $F(z) = R \in \mathbb{R}$ with $\operatorname{Im} z > 0$, then $\operatorname{Im}(R/z) = -R \operatorname{Im} z / |z|^2 > 0$, so $R < 0$. $\square$

Corollary 5.3 (Gaussian negativity and strict log-convexity). *For every $a, b > 0$,*

$$-g_{a,b} = \int_{0 < y < x < 1} W_{a,b}(x, y) \frac{x+y}{(1+x^2)(1+y^2)} dy dx > 0, \quad (5.4)$$

and

$$\frac{H_2^{(b)}}{4 \cdot 3^a} < -g_{a,b} < \frac{H_2^{(b)}}{3^a}. \quad (5.5)$$

The function $(a, b) \mapsto \Gamma(a)\Gamma(b)(-g_{a,b})$ is strictly log-convex on the positive quadrant.

Proof. Set $z = i$ in (5.1) and rationalize the denominator. Since $1 < (1+x^2)(1+y^2) < 4$ and $\int W(x+y) = c_3 = H_2^{(b)}/3^a$, the bounds follow. For log-convexity apply Hölder's inequality to the positive integral with the gamma factors removed. Its parameter dependence is the exponential of $(a-1)\log(-\log x) + (b-1)\log \log(x/y)$. Equality in Hölder would require a nontrivial real linear combination of these two logarithms to be constant almost everywhere on the open triangle, which is impossible. This proves strictness for every nontrivial line segment of parameter pairs. $\square$

5.1 All-order integral identities

Theorem 5.4 (Hurwitz–polylogarithm transport). *For $a, b > 0$, $b \neq 1$, and $z \in \mathbb{C} \setminus [1, \infty)$,*

$$\begin{aligned} F_{a,b}(z) &= \zeta(b) \operatorname{Li}_a(z) - \frac{\operatorname{Li}_{a+b-1}(z)}{b-1} \\ &\quad - \frac{1}{\Gamma(b)} \int_0^\infty t^{b-1} h(t) \operatorname{Li}_a(ze^{-t}) dt. \end{aligned} \quad (5.6)$$

At $b = 1$ the pole-free identity is

$$F_{a,1}(z) = \gamma \operatorname{Li}_a(z) - \partial_a \operatorname{Li}_a(z) - \int_0^\infty h(t) \operatorname{Li}_a(ze^{-t}) dt. \quad (5.7)$$

In particular, the index $a + b - 1$ in (5.6) is allowed to be zero or negative.

Proof. Inside the disk, multiply (2.3), with $q = n$, by $n^{-a}z^n$ and sum. Exponential decay at infinity, boundedness of h , and $b > 0$ justify termwise integration. The Hurwitz difference is $H_{n-1}^{(b)}$, so coefficient comparison gives (5.6). At $b = 1$, use $H_{n-1} = \gamma + \log n - \int_0^\infty e^{-nt} h(t) dt$; the sum of $(\log n)n^{-a}z^n$ is $-\partial_a \text{Li}_a(z)$. The integral expressions on the right are locally holomorphic on the slit plane. Near $t = 0$, $\text{Li}_a(ze^{-t})$ stays bounded on compact slit sets, and at infinity it is $O(e^{-t})$. Analytic continuation therefore proves the identities throughout the claimed domain. Standard conventions for Li_s are those of DLMF 25.12 [4]. $\square$

These identities are useful even when no signed Hausdorff measure exists. They are transformations to a depth-one polylogarithm integral, not assertions that a depth-two constant has been reduced to a finite basis of depth-one periods.

6 Slit-plane nonvanishing and the unique angular zero

Lemma 6.1 (Ordered-sign test). *Suppose a nonzero finite signed measure μ has total mass zero, its negative part is supported to the left of r , and its positive part to the right, allowing an atom at r in either part. If q is strictly increasing on the relevant supports and integrable against $|\mu|$, then $\int q d\mu > 0$. For strictly decreasing q the inequality is reversed. These conclusions apply both to the one-crossing density of Theorem 3.1 and to its critical endpoint measure.*

Proof. Subtract $q(r)\mu([0, 1]) = 0$. On each nonzero signed part, $(q(u) - q(r)) d\mu(u)$ has the asserted sign. The strict positive density on an interior interval ensures that the integral is strict, even when the other part is concentrated at the endpoint $r = 1$. $\square$

Theorem 6.2 (Zero-free principal branch on the full closed domain). *For $a, b > 0$ with $w \geq 1$, $F_{a,b}$ has exactly one zero on $\mathbb{C} \setminus [1, \infty)$: its double zero at the origin. More explicitly, let r be the crossing of Theorem 3.1 for $w > 1$, and let $r = 1$ for $w = 1$. Then*

$$F_{a,b}(z) = \frac{z^2}{1 - rz} P_{a,b}(z), \quad (6.1)$$

$$P_{a,b}(z) = \int_{[0,1]} \frac{u - r}{1 - zu} d\mu_{a,b}(u). \quad (6.2)$$

The measure $(u - r) d\mu_{a,b}(u)$ is positive and nonzero. The imaginary part of $P_{a,b}(z)$ has the strict sign of $\text{Im } z$, and $P_{a,b}(x) > 0$ for real $x < 1$.

Proof. The moment representation gives $F(z) = \int z(1 - zu)^{-1} d\mu(u)$, first in the disk and then on the slit plane. Subtract $(1 - rz)^{-1} \int z d\mu = 0$ to obtain (6.1). Positivity of $(u - r) d\mu$ follows from the sign pattern; on the critical line the atom disappears after multiplication by $u - 1$, leaving $(1 - u) d\nu(u) > 0$. For nonreal z ,

$$\text{Im } P(z) = (\text{Im } z) \int \frac{u(u - r)}{|1 - zu|^2} d\mu(u),$$

and the integral is strictly positive. For real $z < 1$ all denominators in (6.2) are positive. Thus P never vanishes there, and $1 - rz$ has no zero off the excluded cut. Finally $P(0) = \int u d\mu(u) = c_2 = 2^{-a}$, proving exact multiplicity two. $\square$

The factorization mechanism is already used in the angular continuation for $a \geq 1$ [2]. The substantive extension here is the construction of the required signed measure for $0 < a < 1, w \geq 1$, including its atomic endpoint.

Theorem 6.3 (Unique simple angular zero). *Let $a, b > 0$, $w \geq 1$, and $0 < \rho \leq 1$. There is exactly one $\theta_{a,b}(\rho) \in (0, \pi)$ satisfying $\operatorname{Im} F_{a,b}(\rho e^{i\theta}) = 0$. It obeys*

$$\arccos \rho < \theta_{a,b}(\rho) < \frac{\pi}{2}. \quad (6.3)$$

The imaginary part is positive before this zero and negative after it, and its angular derivative at the zero is strictly negative. The zero depends smoothly on the radius, including a one-sided interpretation at $\rho = 1$.

Proof. Set $c = \cos \theta$ and

$$K_{\rho,c}(u) = \frac{1}{1 - 2\rho cu + \rho^2 u^2}, \quad J(\rho, c) = \int K_{\rho,c}(u) d\mu(u).$$

Then $\operatorname{Im} F(\rho e^{i\theta}) = \rho \sin \theta J(\rho, c)$. The weight $K_{\rho,0}$ is strictly decreasing, so $J(\rho, 0) < 0$. For $\rho < 1$, the weight $K_{\rho,\rho}$ is strictly increasing on $(0, 1)$, since its derivative has the sign of $1 - u$. Thus $J(\rho, \rho) > 0$, providing a zero $0 < c < \rho$.

For $\rho = 1$ and $w > 1$, the negative measure is supported away from $u = 1$. Separating the two signs before expanding the endpoint kernel gives

$$\lim_{c \uparrow 1} J(1, c) = \int \frac{d\mu(u)}{(1 - u)^2} = \sum_{n \geq 1} n c_n > 0,$$

with $+\infty$ allowed. The negative part is integrable at this endpoint and monotone convergence applies to the positive part. On $w = 1$ that particular argument must not be used: the negative density also reaches one. Instead write $\mu = M\delta_1 - \nu$. Then

$$(1 - c)J(1, c) = \frac{M}{2} - \int \frac{1 - c}{1 - 2cu + u^2} d\nu(u) \longrightarrow \frac{M}{2} > 0.$$

Indeed the last integrand tends to zero at every $u < 1$ and is bounded by one for $0 < c < 1$, because its denominator is at least $1 - c^2 \geq 1 - c$. The finite measure ν has no atom at one. This proves existence at unit radius as well.

For $c' > c$,

$$\frac{d}{du} \frac{K_{\rho,c'}(u)}{K_{\rho,c}(u)} = \frac{2\rho(c' - c)(1 - \rho^2 u^2)}{(1 - 2\rho c' u + \rho^2 u^2)^2} > 0 \quad (0 < u < 1). \quad (6.4)$$

At a zero $J(\rho, c) = 0$, the measure $K_{\rho,c} d\mu$ has zero mass and the same ordered signs. Lemma 6.1, applied to this ratio, gives $J(\rho, c') > 0$ for every $c' > c$, and the reversed inequality for every $c' < c$. This proves uniqueness and the global sign pattern. At the zero,

$$J_c = \int K_{\rho,c}(u) \frac{2\rho u}{1 - 2\rho cu + \rho^2 u^2} d\mu(u) > 0,$$

since the second factor is strictly increasing on $(0, 1)$. Consequently the angular derivative is $-\rho \sin^2 \theta J_c < 0$. All differentiations are dominated away from the point $z = 1$, and the implicit function theorem gives smooth radial dependence. $\square$

By Theorem 4.2 and the strict sign brackets in this proof, for every fixed $0 < \rho \leq 1$ and $0 < b < 1$,

$$\theta_{1-b+\varepsilon,b}(\rho) \longrightarrow \theta_{1-b,b}(\rho).$$

Thus the atom is compatible with continuous angular geometry; it changes the representing measure and summation interpretation, not the existence of a well-defined limiting root.

7 Normalized-radius monotonicity at opposite endpoints

The normalized radial quantity is

$$\eta_{a,b}(\rho) = \frac{\cos \theta_{a,b}(\rho)}{\rho}. \quad (7.1)$$

The earlier report conjectures its decrease for integer $a \geq 2$ and all $b > 0$, and for $a = 1, b > 1$, but its increase for $a = 1, 0 < b < 1$ [2, Conjecture “Normalized-radius monotonicity”]. No such broad statement follows from uniqueness of the zero. We prove two endpoint-measure theorems that isolate an exact mechanism for the two directions.

7.1 A right-endpoint atom

Theorem 7.1 (Critical normalized-radius theorem). *Let $a = 1 - b \in (0, 1)$. The function $\eta_{a,b}(\rho)$ is strictly increasing for $0 < \rho \leq 1$, and*

$$\frac{1}{2} < \eta_{a,b}(\rho) < 1, \quad \arccos \rho < \theta_{a,b}(\rho) < \arccos(\rho/2). \quad (7.2)$$

The following positive integral gives its derivative. Put $x = \rho^2$, $m = 2\eta - 1$, $d\omega(u) = (1 - u) d\nu_{a,b}(u)$, and $D(u) = 1 + x\{u^2 - (1 + m)u\}$. Then $0 < m < 1$ and

$$\frac{dm}{dx} = \frac{\int_0^1 \frac{(u - m)^2(1 - u)}{D(u)^2} d\omega(u)}{(1 - xm) \int_0^1 \frac{1 - xu}{D(u)^2} d\omega(u)} > 0. \quad (7.3)$$

In particular $\eta'(\rho) = \rho m'(\rho^2) > 0$.

Proof. We prove the identity for any measure $M\delta_1 - \nu$ with $\nu([0, 1]) = M$ and strictly positive density on $(0, 1)$, which includes (3.4). With $c = \rho(1 + m)/2$, $D(1) = 1 - xm$ and

$$J(\rho, c) = \frac{M}{D(1)} - \int \frac{d\nu(u)}{D(u)} = \frac{x}{1 - xm} G(m, x),$$

where

$$G(m, x) = \int \frac{m - u}{D(u)} d\omega(u). \quad (7.4)$$

Here we used $D(u) - D(1) = x(u - 1)(u - m)$. For fixed $0 < x \leq 1$, $G(0, x) < 0$ and $G(1, x) > 0$. At $x = 1, m = 1$, the latter integrand simplifies to $d\nu(u)$, so it is still integrable and positive. It is also the left limit as $m \uparrow 1$: for $1/2 \leq m < 1$, the corresponding scalar integrand against ν lies between -1 and 1 , so dominated convergence applies. For interior $0 < m < 1$,

$$G_m = \int \frac{1 - xu}{D(u)^2} d\omega(u) > 0. \quad (7.5)$$

Thus its root lies strictly in $(0, 1)$ and coincides with the angular root already proved unique.

Let $d(u) = u^2 - (1 + m)u$. At a root of G , the probability measure proportional to $d\omega(u)/D(u)$ has mean m . The elementary identities

$$d(m) = -m, \quad d(u) - d(m) = (u - m)(u - 1)$$

therefore give

$$\begin{aligned} G_x &= \int (u - m) \frac{d(u)}{D(u)} \frac{d\omega(u)}{D(u)} \\ &= \int (u - m) \left(\frac{d(u)}{D(u)} - \frac{-m}{1 - xm} \right) \frac{d\omega(u)}{D(u)} \\ &= -\frac{1}{1 - xm} \int \frac{(u - m)^2(1 - u)}{D(u)^2} d\omega(u) < 0. \end{aligned}$$

The middle subtraction integrates to zero. Combining $m' = -G_x/G_m$ with (7.5) proves (7.3). All denominators are bounded away from zero locally at the root, including at $x = 1$ because $m < 1$. Strictness follows from the nondegenerate positive density. $\square$

Corollary 7.2 (A positive small-radius coefficient). *Let $a = 1 - b \in (0, 1)$ and abbreviate $A = H_2^{(b)}$, $B = H_3^{(b)}$, $C = H_4^{(b)}$. Then*

$$\begin{aligned} \eta_{a,b}(\rho) &= \frac{A}{2} \left(\frac{2}{3} \right)^a \\ &+ \left\{ AB \left(\frac{1}{3} \right)^a - \frac{C}{2} \left(\frac{2}{5} \right)^a - \frac{A^3}{2} \left(\frac{8}{27} \right)^a \right\} \rho^2 + O(\rho^4). \end{aligned} \quad (7.6)$$

The expression in braces is strictly positive on this critical line. More intrinsically, put $m_0 = (c_3 - c_2)/c_2$ and let U have probability measure $d\omega/c_2$. That coefficient equals

$$\frac{1}{2} \mathbb{E}[(U - m_0)^2(1 - U)] > 0. \quad (7.7)$$

Proof. At $x = 0$, $G(m, 0) = 0$ gives $m_0 = \int u d\omega / \int d\omega$. The moments of ω are

$$\int u^j d\omega = D_{j+1} - D_{j+2} = c_{j+2} - c_{j+1}.$$

Thus its mass is c_2 and $(1 + m_0)/2 = c_3/(2c_2) = A(2/3)^a/2$. Formula (7.3) at $x = 0$ gives the coefficient (7.7). Expressing its moments through c_2, c_3, c_4, c_5 and simplifying gives

$$\frac{c_3 c_4}{c_2^2} - \frac{c_5}{2c_2} - \frac{c_3^3}{2c_2^3},$$

which is exactly the braces in (7.6). Analytic implicit dependence near $x = 0$ gives the remainder. The formal expansion itself is already present for $a \geq 1$ in the angular report; the positive variance formula and the strict sign on $a + b = 1$ are the additional conclusions here. $\square$

7.2 A dual left-endpoint atom

Theorem 7.3 (Opposite monotonicity for a left atom). *Suppose $\mu = \nu - M\delta_0$, where ν has strictly positive integrable density on $(0, 1)$ and mass M . Define $F(z) = \int z(1 - zu)^{-1} d\mu(u)$. On every upper semicircle of radius $0 < \rho \leq 1$, its imaginary part has one simple zero, and its normalized displacement $\eta(\rho) = \cos \theta(\rho)/\rho$ satisfies $0 < \eta < 1/2$ and $\eta'(\rho) < 0$.*

More explicitly, set $x = \rho^2$, $m = 2\eta$, $d\omega = u d\nu$, and $D(u) = 1 + x(u^2 - mu)$. At the root,

$$\frac{dm}{dx} = - \frac{\int \frac{u(u-m)^2}{D(u)^2} d\omega(u)}{\int \frac{1}{D(u)^2} d\omega(u)} < 0. \quad (7.8)$$

Proof. In this case

$$J(\rho, c) = \int \frac{d\nu(u)}{D(u)} - M = xG(m, x), \quad G(m, x) = \int \frac{m-u}{D(u)} d\omega(u).$$

The function G is negative at $m = 0$, positive at $m = 1$, and

$$G_m = \int D(u)^{-2} d\omega(u) > 0, \quad G_x = \int u(u-m)^2 D(u)^{-2} d\omega(u) > 0.$$

The denominators stay positive for $0 \leq m \leq 1$ and $0 \leq x \leq 1$. This proves the unique simple root, its range, and (7.8). The signs outside the root also follow from the same increasing-kernel-ratio argument as before. $\square$

Corollary 7.4 (A polylogarithm-minus-linear-term family). *For every real $a > 0$, the normalized angular displacement of $\text{Li}_a(z) - z$ decreases strictly with $\rho \in (0, 1]$ and belongs to $(0, 1/2)$. This family is the limit of $F_{a,b}$ as $b \rightarrow \infty$.*

Proof. The probability measure $d\nu(u) = k_a(-\log u) du$ has moments n^{-a} . Thus $\nu - \delta_0$ has moments 0 at $n = 1$ and n^{-a} for $n \geq 2$, which gives $\text{Li}_a(z) - z$. Apply Theorem 7.3. Coefficient convergence gives the indicated limit in the disk. For sufficiently large $b > 1$, the signed measures of Theorem 3.1 have uniformly bounded variation, as proved below. Moment uniqueness and polynomial approximation then give weak convergence to $\nu - \delta_0$, hence locally uniform convergence on the slit plane. $\square$

These two theorems do not settle the finite- b normalized-radius conjecture from the baseline. They show why the orientation of an endpoint atom can force opposite directions, and supply rigorous boundary data for a continuous-parameter phase diagram.

8 Elementary critical-kernel identities and Hurwitz inequalities

In this section $0 < b < 1$, $a = 1 - b$, and $C_b(T) = C_{1-b,b}(T) = \mathbb{E}[h(TV)]$ with $V \sim \text{Beta}(b, 1 - b)$. In particular $1/2 < C_b(T) < 1$ and C_b is strictly increasing.

Theorem 8.1 (An elementary convergent series). *For every $T > 0$,*

$$C_b(T) = \frac{1}{2} + \sum_{k=1}^{\infty} \frac{\sin(b \arctan(T/(2\pi k)))}{\pi k (1 + (T/(2\pi k))^2)^{b/2}}. \quad (8.1)$$

Every summand is positive. The tail after $K \geq 1$ terms is bounded by

$$0 < C_b(T) - \frac{1}{2} - \sum_{k=1}^K (\text{summand}) \leq \frac{bT}{2\pi^2 K}. \quad (8.2)$$

For $|T| < 2\pi$ it also has the convergent expansion

$$C_b(T) = \frac{1}{2} + \sum_{j=1}^{\infty} \frac{B_{2j}(b)_{2j-1}}{(2j)!(2j-1)!} T^{2j-1}, \quad (8.3)$$

where $(b)_m$ is the rising factorial and B_{2j} is a Bernoulli number, not a Bernoulli polynomial evaluated at b .

Proof. The classical partial fraction for $\coth$ gives

$$h(t) = \frac{1}{2} + 2t \sum_{k=1}^{\infty} \frac{1}{t^2 + (2\pi k)^2}. \quad (8.4)$$

This follows from DLMF 4.36.3 after rescaling [5]. The summands are positive for $t > 0$, so Tonelli permits averaging over V . For $|z| < 1$, the beta moments yield

$$\mathbb{E} \frac{1}{1 - zV} = \sum_{j \geq 0} z^j \frac{(b)_j}{j!} = (1 - z)^{-b}.$$

Both sides are analytic for $z \in \mathbb{C} \setminus [1, \infty)$, so the same identity holds there. At $z = iq$, its imaginary part is

$$\mathbb{E} \frac{qV}{1 + q^2 V^2} = \frac{\sin(b \arctan q)}{(1 + q^2)^{b/2}}.$$

Taking $q = T/(2\pi k)$ proves (8.1). Since $\sin(b \arctan q) \leq bq$ and the denominator factor is at least one, its tail is at most $bT(2\pi^2)^{-1} \sum_{k > K} k^{-2} \leq bT/(2\pi^2 K)$.

The usual Bernoulli expansion of (2.1) has radius 2π : $h(t) = 1/2 + \sum_{j \geq 1} B_{2j} t^{2j-1}/(2j)!$. For $|T| < 2\pi$ it is uniformly absolutely convergent for $0 \leq V \leq 1$. Average termwise and use $\mathbb{E} V^{2j-1} = (b)_{2j-1}/(2j-1)!$ to obtain (8.3). $\square$

For instance,

$$C_{1/2}(T) = \frac{1}{2} + \frac{T}{24} - \frac{T^3}{2304} + \frac{T^5}{122880} + O(T^7). \quad (8.5)$$

The companion program regenerates exact coefficients through degree 13 at $b = 1/10, 1/2, 9/10$ by two algebraically equivalent constructions. The numerical tests of (8.1) use independent beta quadrature, with the analytic tail bound displayed separately.

Theorem 8.2 (A completely monotone normalized Hurwitz expression). *For $0 < b < 1$ and $q > 0$, define*

$$L_b(q) = q^{b-1} \zeta(b, q) + \frac{1}{1-b}. \quad (8.6)$$

Then

$$L_b(q) = \int_0^\infty e^{-qT} C_b(T) dT, \quad \frac{1}{2q} < L_b(q) < \frac{1}{q}. \quad (8.7)$$

In particular, $(-1)^j L_b^{(j)}(q) > 0$ for all $j \geq 0$. The function

$$qL_b(q) = q^b \zeta(b, q) + \frac{q}{1-b} \quad (8.8)$$

decreases strictly from 1 to $1/2$ as q increases from zero to infinity.

Proof. Formula (8.7) is (2.8) with $a = 1 - b$. Bounds and strict complete monotonicity follow from $1/2 < C_b < 1$ and differentiating the positive Laplace integral. Substitution $T = s/q$ gives

$$qL_b(q) = \int_0^\infty e^{-s} C_b(s/q) ds.$$

Strict increase of C_b proves strict decrease with q . The endpoint limits follow by dominated convergence and the limits $C_b(0+) = 1/2$, $C_b(\infty) = 1$. $\square$

The beta average is the common source of the elementary series and the Hurwitz inequalities. The inequalities, finite-difference signs, and Hankel determinants are analytic consequences of a positive integral, not numerical pattern extrapolations.

9 Euler enclosures and the failure of an unrestricted constant one

Let $a, b > 0$ with $w \geq 1$. Define

$$f(n) = c_{2n+1} = \frac{H_{2n}^{(b)}}{(2n+1)^a}, \quad d_k = \sum_{n=0}^k (-1)^n \binom{k}{n} f(n), \quad E_N = \sum_{k=0}^{N-1} \frac{d_k}{2^{k+1}}. \quad (9.1)$$

The error is measured against $g_{a,b} = \text{Im } F_{a,b}(i)$. On the critical line this is an Abel value; the ordinary alternating series with terms $(-1)^n f(n)$ diverges by the term test.

Theorem 9.1 (Extended one-sided Euler bounds). *For every $k \geq 1$ one has $d_k < 0$, while $d_0 = 0$. The E_N decrease strictly to $g_{a,b}$, and*

$$E_N - g_{a,b} = -2^{-N} \int_{[0,1]} \frac{(1-u^2)^N}{1+u^2} d\mu_{a,b}(u) > 0. \quad (9.2)$$

Consequently

$$E_N - M(a,b)2^{-N} < g_{a,b} < E_N \quad (N \geq 1), \quad (9.3)$$

with the following elementary choices:

$$M(a,b) = \begin{cases} 1, & a \geq 1, \\ (1-b)^{-1}, & 0 < a < 1, \ 0 < b < 1, \ w \geq 1, \\ b/(b-1), & 0 < a < 1, \ b > 1, \\ 1 + 1/a, & 0 < a < 1, \ b = 1. \end{cases} \quad (9.4)$$

The choice one in the first line is the prior optimal universal constant on that restricted domain; it is not asserted to work in the other lines.

Proof. The binomial theorem and (1.2) give $d_k = \int (1 - u^2)^k d\mu(u)$. Thus $d_0 = 0$, while the strictly decreasing weight gives $d_k < 0$ for $k \geq 1$. With $m = m(\mu)$, we have $|d_k| \leq m$. Therefore the Euler series is absolutely convergent and may be summed under the measure integral:

$$\sum_{k \geq 0} \frac{d_k}{2^{k+1}} = \int \frac{d\mu(u)}{1 + u^2} = \operatorname{Im} F(i).$$

Summing the geometric tail gives (9.2). Its weight is strictly decreasing, bounded by one, and strictly less than one for every $u > 0$. Since the negative Jordan part has no atom at zero, $0 < E_N - g < m2^{-N}$. It remains to bound m .

For $b < 1, w > 1$, rewrite (3.2) as

$$K_{a,b} = \frac{k_{w-1}}{1-b} - [-\zeta(b)k_a + C_{a,b}].$$

The two displayed densities are positive, and each has mass $1/(1-b)$ in the u measure, by zero total mass. Hence $m < 1/(1-b)$. On $w = 1$ the exact Jordan mass is $1/a = 1/(1-b)$. For $b > 1$, (3.2) is a difference of positive densities $\zeta(b)k_a$ and $k_{w-1}/(b-1) + C_{a,b}$, so $m < \zeta(b) < 1 + 1/(b-1) = b/(b-1)$.

For $b = 1$ and $0 < a < 1$, use a translation estimate rather than the singular limit of either of these bounds. Let σ_a be the probability law of $u = e^{-Y}$ with $Y \sim \text{Gamma}(a, 1)$. Its logarithmic density $p_a(s) = e^{-s}s^{a-1}/\Gamma(a)$ decreases strictly. If $D_{e^{-t}}$ denotes pushforward under $u \mapsto e^{-t}u$, then direct comparison of $p_a(s)$ and $p_a(s-t)$ gives

$$\|\sigma_a - D_{e^{-t}}\sigma_a\|_{\text{TV}} = 2 \int_0^t p_a(s) ds \leq \frac{2t^a}{\Gamma(a+1)}.$$

The signed-measure integral

$$\tilde{\mu} = \int_0^\infty \frac{\sigma_a - D_{e^{-t}}\sigma_a}{e^t - 1} dt$$

converges in total variation. Its $(n-1)$ st moment is $n^{-a} \int_0^\infty (1 - e^{-(n-1)t})(e^t - 1)^{-1} dt = c_n$. By uniqueness it is $\mu_{a,1}$, and

$$m \leq \frac{1}{\Gamma(a+1)} \int_0^\infty \frac{t^a}{e^t - 1} dt = \zeta(a+1) < 1 + 1/a.$$

Finally, for $a \geq 1$ the earlier sharper bound can be recovered without any fractional-order gap. At $a = 1$ the equivalent base density is

$$K_{1,b}(T) = -\frac{T^b}{\Gamma(b+1)} + \frac{1}{\Gamma(b)} \int_T^\infty \frac{t^{b-1}}{e^t - 1} dt.$$

Its n th Laplace value is $-n^{-b-1} + n^{-1} \sum_{j=1}^n j^{-b} = H_{n-1}^{(b)}/n$, by Tonelli and a finite geometric sum. Moment uniqueness therefore identifies this base density with (3.2) or (3.3). Both positive terms in this difference have $e^{-T} dT$ integral one. For the second term, interchange the integrals and use $(1 - e^{-t})/(e^t - 1) = e^{-t}$; the first is a gamma integral. Their positive overlap makes the Jordan mass strictly less than one. For $a > 1$, convolution by k_{a-1} in the T coordinate preserves moments after multiplication by $n^{-(a-1)}$ and contracts the $L^1(e^{-T} dT)$ norm, because $\int_0^\infty e^{-T} k_{a-1}(T) dT = 1$. Moment uniqueness identifies the result with the present kernel, so $m < 1$ for every real $a \geq 1$. This proves all four choices. $\square$

The same translation proof gives, when $0 < a \leq 1$ and $a + b > 1$, the optional non-elementary bound

$$m \leq \frac{\Gamma(a+b)\zeta(a+b)}{\Gamma(a+1)\Gamma(b)}.$$

It may improve a particular estimate, but diverges as $a + b \downarrow 1$. The critical atomic construction is therefore not replaced by this translation argument.

Proposition 9.2 (Exact finite Euler weights). *For $N \geq 1$,*

$$E_N = 2^{-N} \sum_{n=0}^{N-1} (-1)^n f(n) \sum_{j=n+1}^N \binom{N}{j}. \quad (9.5)$$

Thus the finite sum can be formed without constructing a triangular difference table.

Proof. Substitute the definition of d_k and interchange the finite sums. The coefficient identity is

$$\sum_{k=n}^{N-1} 2^{N-k-1} \binom{k}{n} = \sum_{j=n+1}^N \binom{N}{j},$$

which follows by induction from Pascal's identity. Harmonic numbers and binomial tails can then each be updated successively. This rearrangement is already used in the integer-order manuscript; the new feature in the implementation here is certified rational interval arithmetic for fractional powers. $\square$

Theorem 9.3 (A certified scope obstruction). *For $a = 1/10$ and $b = 9/10$,*

$$\begin{aligned} -0.5297619322857712630946293655 &< g_{1/10,9/10} \\ &< -0.5297619322857712630946293653. \end{aligned} \quad (9.6)$$

In particular, $g_{1/10,9/10} < -13/25 < -1/2$. Since $E_1 = 0$, the inequality $E_N - 2^{-N} < g_{a,b}$ fails at $N = 1$ for this pair. A uniform constant one cannot be extended to the entire region $a + b \geq 1$. The failure also occurs strictly above the threshold, with ordinary boundary convergence:

$$\begin{aligned} -0.5308442529661593261720712602 &< g_{1/10,1} \\ &< -0.5308442529661593261720712600 < -0.53. \end{aligned} \quad (9.7)$$

Proof. The accompanying standard-library program `code/certify.py` evaluates (9.5) with $N = 96$ by outward rational interval arithmetic. Every fractional power is bounded by integer tenth-root inequalities, as detailed in Section 10. The critical mass bound is the exact rational $M = 10$. Subtracting $10 \cdot 2^{-96}$ from the lower endpoint of the finite Euler enclosure and leaving its upper endpoint unchanged gives the interval recorded in `data/exact_certificates.json`. Outward decimal rounding yields (9.6); exact rational comparisons verify its upper endpoint is less than $-13/25$. No numerical polylogarithm, floating-point root, or unproved identity enters this certificate. For $(a, b) = (1/10, 1)$ the same computation uses $M = 1 + 1/a = 11$ and proves (9.7). Its ordinary boundary convergence follows from $a + b = 11/10 > 1$ and Corollary 3.3. $\square$

This is *not* a counterexample to the baseline theorem, which explicitly assumes $a \geq 1$. It is a counterexample to dropping that hypothesis while retaining the same universal error constant.

Proposition 9.4 (The exact critical constant for fixed parameters). *On $a + b = 1$, the scaled errors $2^N(E_N - g_{a,b})$ decrease strictly to zero for $N \geq 1$. The smallest constant in the non-strict estimate $E_N - g_{a,b} \leq C2^{-N}$, valid for all integers $N \geq 1$ at that fixed pair, is $C = -2g_{a,b}$.*

Proof. The endpoint atom contributes zero to (9.2) when $N \geq 1$. Hence

$$2^N(E_N - g) = \int_0^1 \frac{(1 - u^2)^N}{1 + u^2} d\nu(u).$$

Strict pointwise decrease and dominated convergence prove the first claim. Its largest value occurs at $N = 1$ and is $2(E_1 - g) = -2g$. For the corresponding strict inequality the infimum of admissible constants is the same, but equality at $N = 1$ prevents attaining it. $\square$

10 Exact certificates and independent diagnostics

10.1 Certified fractional powers

For a positive rational exponent p/q and an integer $n \geq 1$, set $B = 10^{65}$ and compute the integer $r = \lfloor (n^p B^q)^{1/q} \rfloor$. The verifier explicitly checks

$$r^q \leq n^p B^q < (r + 1)^q. \quad (10.1)$$

If equality holds on the left, $n^{-p/q} = B/r$ exactly. Otherwise

$$\frac{B}{r + 1} < n^{-p/q} < \frac{B}{r}. \quad (10.2)$$

All operations in these statements are integer or rational. The integer Newton iteration is merely a method to propose r ; (10.1), not its convergence history, validates the result. Additions, sign-aware scalar products, and positive products are rounded outward onto the grid $10^{-60}\mathbb{Z}$ using integer division. This keeps numerator sizes manageable without introducing floating-point assumptions.

Finite Euler centers at fractional indices are generally algebraic, not rational. The program encloses them by rational intervals, then adds the proved rational tail. It does not mislabel an irrational finite center as an exact rational value.

a	b	$g_{a,b}$, displayed approximately
1/10	9/10	-0.52976193228577126309
1/10	1	-0.53084425296615932617
1/2	1/2	-0.40393698771701357983
3/4	1/2	-0.33913987183657185082
1/2	1	-0.40169999261081928166
1/4	3/2	-0.47502596533036419882
1	1	-0.27219826128795026631
2	3	-0.09324795037117472982

These displayed approximations are summaries, not the certified endpoints. The JSON stores outward lower and upper decimal bounds for each of the eight $N = 96$ enclosures, with widths and mass bounds. For (1, 1) the value agrees with the known $-\pi \log 2/8$; that identity is a convention check, not a premise of the fractional-order counterexample.

10.2 Angular certificates without trigonometric rounding

Let $0 < \rho < 1$ and $-1 < c < 1$ be rational. The Chebyshev identity gives

$$\frac{\operatorname{Im} F_{a,b}(\rho e^{i\theta})}{\sin \theta} = \sum_{n \geq 1} c_n \rho^n U_{n-1}(c), \quad c = \cos \theta, \quad (10.3)$$

where $U_{-1} = 0$, $U_0 = 1$, and $U_{n+1} = 2cU_n - U_{n-1}$. Thus every polynomial value is rational. For $|c| < 1$, $|U_{n-1}(c)| \leq n$; this follows from $|\sin(n\theta)| \leq n|\sin \theta|$. Also $c_n \leq n$ for every positive a, b . With $j = N + 1$, a valid explicit tail is therefore

$$\left| \sum_{n > N} c_n \rho^n U_{n-1}(c) \right| \leq \rho^j \left\{ \frac{j^2}{1 - \rho} + \frac{2j\rho}{(1 - \rho)^2} + \frac{\rho(1 + \rho)}{(1 - \rho)^3} \right\}. \quad (10.4)$$

The formula sums $\sum_{n \geq j} n^2 \rho^n$ exactly.

The numerical program proposes rational endpoints for c , but the separate rational verifier accepts a bracket only when the interval in (10.3) is strictly negative at the lower endpoint and strictly positive at the upper endpoint, including (10.4). The theorem supplies uniqueness when $a + b \geq 1$. For the deliberately included test $(a, b, \rho) = (1/4, 1/4, 1/2)$, it certifies only existence of a root in the bracket; no uniqueness conclusion is attached.

a	b	$\eta(1/4)$	$\eta(1/2)$	$\eta(3/4)$
1/10	9/10	0.7387056020	0.7429273646	0.7517703302
1/2	1/2	0.6980111081	0.7015741355	0.7088461987
9/10	1/10	0.6719709290	0.6751086399	0.6814072072

The critical rows illustrate Theorem 7.1; the proof does not infer monotonicity from this table. The underlying rational c -brackets have width 5×10^{-12} . Nine critical brackets, two strictly supercritical brackets, and one subcritical existence bracket are replayed.

10.3 Independent tests and their limits

The recorded run has the following evidence layers.

Exact arithmetic. The standard-library verifier passes all eight Euler enclosures and twelve angular brackets, checking 7,430 integer-root inequalities. A symbolic program verifies nine rational identities underlying the angular kernels, the two derivative formulas, and the small-radius variance reduction. A separate symbolic routine checks 21 exact Bernoulli coefficients at three rational parameters. None of these finite checks certifies the analytic convergence arguments by itself.

Numerical diagnostics. Independent beta–gamma quadrature checks 50 Laplace moments; the largest observed absolute discrepancy in the supplied run is about 2.42×10^{-13} . Positive double quadrature separately evaluates three critical Gaussian values. Twelve elementary-series tests compare the partial sum and its analytic tail with beta quadrature. Twelve critical Laplace-identity tests have largest observed discrepancy about 2.54×10^{-11} . A 15-point critical-constant grid provides conjectural evidence only. These are floating-point diagnostics, not rigorous interval enclosures, and the stated discrepancies are observations, not proved error bounds for the quadratures.

Reproduction. From the package root, with Python and the optional diagnostic dependencies installed, run

```
python code/audit.py
python code/critical_identities.py
python code/certify.py
python code/build_article.py
```

The third command alone replays the exact certificates using only Python’s standard library and the supplied rational root proposals. The first two commands are not prerequisites for trusting the rational endpoint tests, and floating-point root proposals are never accepted without a fresh sign certificate. The build command requires a local LaTeX installation. A separate manifest verifies package bytes; it is an integrity check, not a mathematical proof.

11 Audit conclusions and manuscript integration

11.1 Corrections that an extension actually requires

No error in the *stated* $a \geq 1$ zero-free or Euler-constant-one theorems was established during this targeted audit. The following points are concrete corrections to an unrestricted continuation of them, and should be made explicit when integrating the new results.

The critical kernel is a measure, not just a function. Substituting $a + b = 1$ into a formula valid for $a + b > 1$ and discarding the k_{a+b-1} term would delete the positive atom. The remaining density has mass $-1/a$, not zero, and produces the wrong moments. The correction is exactly $a^{-1}\delta_1$. Corollary 3.2 also proves that no alternative integrable density can repair this.

Boundary convergence changes at the same line. At $a + b = 1$, $c_n \rightarrow 1/a$; the Gaussian alternating terms likewise fail to tend to zero. Statements about ordinary convergence must retain $a + b > 1$. Euler acceleration on the critical line computes the analytic/Abel value, with the measure proof justifying that value. It must not be described as summing an ordinarily convergent series.

The uniform constant one does not survive automatically. Theorem 9.3 proves the obstruction with exact rational arithmetic. The old hypothesis $a \geq 1$ is mathematically substantive. Use the bounds in (9.4), or a separately proved sharper constant, in the extended region.

Fractional integration needs its own sign proof. A generic fractional integral does not come with an automatic one-crossing theorem. The proof here uses the specific monotonicity of K/k_a in (3.5) and (3.6); it does not invoke an unproved variation-diminishing assertion. Similarly, slit-plane zero-freeness below $a + b = 1$ must not be inferred from a finite signed measure that does not exist.

Source status is not reset by a new delivery. The inspected repository has multiple historical S_4 certificates and later integrated proofs. They remain proved prior material. The S_6 identity remains conjectural here; we neither certify it by numerical agreement nor assert its independence from a proposed period basis. The broad normalized-radius conjecture remains open outside the new regimes proved in Section 7.

11.2 Integration map

The suggested destination for this complete package is `Analysis/Polylogarithms/docs/reports/real-order-threshold/`. The main threshold and kernel formulas belong next to `chapters/05-signed-kernels.tex`; the critical endpoint and its root geometry should precede any claim extending the angular report beyond $a \geq 1$. The elementary critical series and Hurwitz inequalities can be cross-referenced from the integration and differentiation chapters.

The supplied integration fragment uses the label prefix `realorder:`. It is an editorial insertion with a summary of the proved results, not a substitute for the complete proofs. The package includes a claim-status ledger, a provenance file with pinned source blob identifiers, and instructions for replaying checks. No upstream source, historical certificate, or repository branch was modified in preparing this delivery.

12 Further research questions and precise conjectures

12.1 The still-open normalized-radius problem

The most direct task is to settle the existing conjecture for integer $a \geq 2$ and finite $b > 0$, together with the $a = 1$ sign transition at $b = 1$. Theorems 7.1 and 7.3 supply opposite endpoint mechanisms but do not interpolate between them. A promising approach is to express the radial derivative at the root as a covariance plus a controlled remainder. The right-atom proof works because $d(u) - d(m)$ factors as $(u - m)(u - 1)$; the left-atom proof works because it factors as $u(u - m)$. For a general one-crossing density no comparable fixed endpoint factor has yet been established.

For real outer orders a richer phase diagram is natural. The critical line has increasing normalized radius, while the $b \rightarrow \infty$ family has decreasing normalized radius for every $a > 0$. A sharp description should locate the transition in (a, b) and decide whether the sign can change with ρ for one fixed pair. The small-radius coefficient in (7.6), valid as a formal local coefficient more generally, gives a necessary local test, but does not prove a global radial sign.

12.2 Below-threshold zeros

Theorem 3.1 rules out finite signed Hausdorff measures when $a + b < 1$, not zero-freeness of the function. The positive double kernel proves nonvanishing in the closed unit disk for all positive orders. It also gives existence of an angular zero on each circle of radius $0 < \rho < 1$: the imaginary part, divided by $\rho^2 \sin \theta$, has integrand

$$W_{a,b}(x, y) \frac{2c - \rho(x + y)}{(1 - 2\rho cx + \rho^2 x^2)(1 - 2\rho cy + \rho^2 y^2)}.$$

At $c = 0$ this is negative and at $c = \rho$ positive. It has fixed sign outside $0 < c < \rho$, so every such zero belongs to that interval. Uniqueness is the additional issue.

Conjecture 12.1 (Angular uniqueness below the moment threshold). *For $a, b > 0$ with $a + b < 1$, the imaginary part of $F_{a,b}$ has exactly one simple zero on each upper semicircle of radius $0 < \rho \leq 1$, with the same sign pattern as in Theorem 6.3.*

The supplied subcritical rational bracket proves existence at one specific parameter triple only. It is not evidence for an exhaustive zero count. At unit radius, even the positive endpoint argument

must be supplied separately; the interior-radius sign test above does not justify an interchange at the singular endpoint $z = 1$.

A second question is whether the slit-plane zero-free theorem holds for all positive a, b . The factor F/z in Corollary 5.2 need not retain a positive imaginary part outside the unit disk: the numerator $1 - xy|z|^2$ can change sign. A different positive representation or a counterexample is needed. We leave this as a question rather than label it a proved extension.

12.3 The optimal uniform critical Euler constant

Proposition 12.2 (Endpoint values for the critical error constant). *On $b = 1 - a$,*

$$\lim_{a \downarrow 0} (-2g_{a,1-a}) = \frac{\pi}{4} + \frac{\log 2}{2}, \quad (12.1)$$

$$\lim_{a \uparrow 1} (-2g_{a,1-a}) = \frac{\pi}{2} - 1. \quad (12.2)$$

Proof. In (5.4), substitute $x = e^{-S/2}$ and $y = xe^{-Y}$. The weight becomes 2^{-a} times the product of independent $\text{Gamma}(a, 1)$ and $\text{Gamma}(b, 1)$ probability densities for S, Y . The remaining bounded continuous function is $(x + y)/((1 + x^2)(1 + y^2))$. As $a \downarrow 0$, $S \rightarrow 0$ in probability and Y converges in law to an exponential variable. Thus $x \rightarrow 1$ and y is asymptotically uniform on $(0, 1)$, giving

$$\lim(-g) = \int_0^1 \frac{1 + y}{2(1 + y^2)} dy = \frac{\pi}{8} + \frac{\log 2}{4}.$$

As $a \uparrow 1$, $Y \rightarrow 0$, while x has limiting density $2x$ on $(0, 1)$. Therefore

$$\lim(-g) = 2 \int_0^1 \frac{x^2}{(1 + x^2)^2} dx = \frac{\pi}{4} - \frac{1}{2}.$$

Weak convergence and boundedness justify both limits, proving the formulas after multiplication by two. $\square$

Conjecture 12.3 (Sharp critical constant). *The function $a \mapsto -2g_{a,1-a}$ decreases strictly on $(0, 1)$. Consequently the sharp constant uniform over the entire critical line and all $N \geq 1$ would be*

$$C_{\text{critical}} = \frac{\pi}{4} + \frac{\log 2}{2}. \quad (12.3)$$

The 15-point diagnostic grid is consistent with this statement. The endpoint limits and Proposition 9.4 prove that any such uniform constant is at least the value in (12.3); the matching global upper bound is not proved in this article. The stronger strict monotonicity is also explicitly conjectural.

12.4 More precise critical computation

The elementary series (8.1) has a simple positive $O(K^{-1})$ tail. Its summands have expansions in inverse even powers of k , so subtracting a finite number of those terms and restoring zeta tails should produce arbitrarily accelerated, sign-controlled versions. An exact implementation should retain outward bounds for trigonometric and fractional-power factors rather than certify a floating-point result after the fact. The Bernoulli expansion is convergent only for $|T| < 2\pi$ and should not be used as a global large- T expansion.

The sharp N -dependence of critical Euler errors is another concrete target. Formula (9.2) localizes at $u = 0$, where the negative density has both a gamma-logarithmic term and a beta-averaged term. Uniform asymptotics as $a \downarrow 0$ would have to resolve competition between N and a , not merely fix the parameters before letting N tend to infinity.

12.5 Higher depth, affine steps, and formal proof

A natural extension replaces $H_{n-1}^{(b)}$ by higher-depth nested harmonic sums. The coefficient growth suggests candidate total-order thresholds, but growth alone proves only nonexistence on one side; it does not produce a one-crossing signed measure on the other. Endpoint atoms may be replaced by several ordered signed components, so angular uniqueness may need to be replaced by a finite zero-count statement.

The manuscript's affine coefficients $H_{rn}^{(b)}/(dn+1)^a$ are another useful direction. Their existing theorem covers $a \geq 1$. A sharp fractional-order threshold should depend on the growth exponent, but the relative scales r, d can affect the kernel sign geometry and hence the best Euler constants. The cancellation between a gamma endpoint and a Hurwitz tail needs to be redone rather than assumed to survive unchanged.

Finally, the proof has a reasonably small formal core: gamma and beta integrals, the regularized Hurwitz identity, polynomial density, ordered-sign measures, and the two rational derivative factorizations. A Lean formalization of the generic endpoint-atom root lemmas would be reusable independently of polylogarithms. The current rational verifier is a replayable finite computation, not such a formalization.

Conclusion

The condition $a \geq 1$ is sufficient but not sharp for the signed harmonic-polylogarithm kernel. The exact finite-measure domain is $a + b \geq 1$, and its boundary carries a mathematically necessary atom. That atom provides both a clean normalized-radius theorem and a warning: ordinary boundary convergence and the old universal Euler constant do not extend unchanged. The positive double kernel, critical Hurwitz identities, and rational fractional-order certificates give separate tools on and below the transition. The remaining conjectures are stated with their proved special cases and their unresolved parts kept distinct.

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